

**This document is a NOTIFIED STANDARD
in the context of the European directives
for interoperability of railway systems**

Railway Group Standard

GE/RT8270

Issue One

Date February 2003

Route Acceptance of Rail Vehicles including changes in Operation or Infrastructure

Synopsis

This standard mandates requirements and responsibilities for the **route acceptance** of rail vehicles for operation on Railtrack controlled infrastructure necessitated by the introduction of a new vehicle, modification to an existing vehicle, change to the operation of the vehicle or change to the infrastructure.

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**Published by:
Railway Safety
Evergreen House
160 Euston Road
London NW1 2DX**

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Part A

A1 Issue record

Issue	Date	Comments
One	February 2003	Original Document Supersedes GO/RT3270 issue two and parts of GC/RT5161 issue two, GE/RT8015 issue one and GE/RT8029 issue one.

This document will be updated when necessary by distribution of a complete replacement.

A2 Implementation of this document

The publication date of this document is 1 February 2003.

This document comes into force on 5 April 2003.

The dates by which compliance with the requirements of this document is to be achieved are set out in Part B2. Where those dates are later than the date on which this document comes into force, this is to give Railway Group members additional time to plan and commence implementation so as to achieve full compliance by the dates set out in Part B2.

This document supersedes the following Railway Group Standards, either in whole or in part as indicated:

Railway Group Standard	Issue No.	Title	RGS sections superseded by this document	Date(s) as of which sections are superseded
GO/RT3270	2	Route Acceptance of Railway Vehicles	All	5 April 2003
GC/RT5161	2	Station Design and Maintenance Requirements	Section 5.1.2 All other sections remain in force	5 April 2003
GE/RT8015	1	Electromagnetic Compatibility between Railway Infrastructure and Trains	Appendix A sections: 1.1.1.1 1.1.1.4 1.1.2.3 1.1.3.1 1.2 All other sections remain in force	5 April 2003
GE/RT8029	1	Management of Clearances and Gauging	Section 14	5 April 2003

GO/RT3270 is withdrawn with effect from 5 April 2003.

GE/RT8029 is withdrawn with effect from 5 April 2003. The remaining parts of GE/RT8029 dealing with defining and maintaining clearances are superseded by [GC/RT5212](#) issue 1 Requirements for Defining and Maintaining Clearances.

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A3 Scope of Railway Group Standards

The overall scope of Railway Group Standards is set out in Appendix A of [GA/RT6001](#). The specific scope of this document is set out in Part B2.

A4 Responsibilities

Railway Group Standards are mandatory on all members of the Railway Group* and apply to all relevant activities that fall into the scope of each individual's Railway Safety Case. If any of those activities are performed by a contractor, the contractor's obligation in respect of Railway Group Standards is determined by the terms of the contract between the respective parties. Where a contractor is a duty holder of a Railway Safety Case then Railway Group Standards apply directly to the activities set out in the Safety Case.

* The Railway Group comprises Railtrack PLC, Railway Safety, and the train and station operators who hold railway safety cases for operation on or related to infrastructure controlled by Railtrack PLC.

Railtrack PLC is known as Railtrack.

A5 Health and safety responsibilities

In issuing this document, Railway Safety makes no warranties, express or implied, that compliance with all or any documents published by Railway Safety is sufficient on its own to ensure safe systems of work or operation. Each user is reminded of its own responsibilities to ensure health and safety at work and its individual duties under health and safety legislation.

A6 Technical content

The technical content of this document is to be approved by:

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A7 Supply

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Part B

B1 Purpose

This document mandates the safety aspects of the process for accepting vehicles to operate on Railtrack controlled infrastructure necessitated by the introduction of a new vehicle, modification to an existing vehicle, change to the operation of the vehicle or change to the infrastructure. It mandates the requirements and responsibilities involved.

The requirements in this standard are separate from requirements for rail vehicles to be approved by HMRI as set out in the Railways and Other Transport Systems (Approval of Works, Plant and Equipment) Regulations 1994.

This standard does not mandate the requirements of The Railways (Interoperability) (High-Speed) Regulations 2002, or the equivalent conventional regulations (when published), but defines the interfaces with those requirements.

B2 Application of this document

B2.1 To whom the requirements apply

This document contains requirements that are applicable to Railway Safety and duty holders of the following categories of Railway Safety Case:

- a) infrastructure controller
- b) train operator
- c) station operator.

This document applies to all routes for which **route acceptance** for a vehicle is required, including those covered by the 'Railways (Interoperability) (High-Speed) Regulations 2002'.

Under the Railways (Safety Case) Regulations 2000, the duty holder at a station (as defined in those Regulations) is responsible for ensuring that the requirements of Railway Group Standards are complied with. At a station, contractual arrangements (including a lease) do not of themselves relieve the duty holder of his obligations under those Regulations.

B2.2 Compliance requirements

B2.2.1 Infrastructure

This document applies to all Railtrack controlled infrastructure over which vehicles operate for scheduled service, planned diversions, movements to depots or workshops, testing and development and any other required occasional movements.

The requirements of this document shall be complied with no later than 5 April 2003.

Where applications for **route acceptance** have already commenced at the date this document comes into force it is permissible for the requirements of the predecessor document to be met. However, where these existing arrangements are to continue beyond a period of six months after this document comes into force the situation shall be regularised through the application for a temporary non-compliance in accordance with [GA/RT6004](#).

B2.2.2 Trains

This document applies to all vehicles intended to operate on Railtrack controlled infrastructure, except those set out in section B2.3.

The requirements of this document shall be complied with no later than 5 April 2003.

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Where applications for **route acceptance** have already commenced at the date this document comes into force it is permissible for the requirements of the predecessor document to be met. However, where these existing arrangements are to continue beyond a period of six months after this document comes into force the situation shall be regularised through the application for a temporary non-compliance in accordance with [GA/RT6004](#).

B2.2.3 Stations

The requirements of this document (section E3.2) shall be complied with no later than 5 April 2003.

B2.2.4 Railway Safety

The requirements of this document (section D5.2) shall be complied no later than 5 April 2003.

B2.2.5 General compliance requirements

Until the compliance date, or the date by which compliance is achieved (if earlier), the applicable requirements of the predecessor documents shall continue to be met (see Part A for details).

After the compliance date, or after the date by which compliance is achieved (if earlier), Railway Group members shall not deviate from the requirements set out in this document.

Where it is considered not reasonably practicable, to comply with the requirements set out in this document, authorisation not to comply shall be sought in accordance with [GA/RT6001](#), [GA/RT6004](#) or [GA/RT6006](#).

B2.3 Exclusions from the application of this document

This document does not apply to:

- a) those shunting vehicles requiring only limited access to Railtrack controlled infrastructure for the specific purpose of transit between sidings; in which case the vehicle and the limited operation shall be separately approved by the infrastructure controller
- b) Road/Rail Vehicles (RRVs) and Rail Mounted Maintenance Machines (RMMs), because these vehicles are only permitted within possessions.

B3 Definitions

Absolute gauging

Absolute gauging of a vehicle is a full assessment of clearances on a section of track between the vehicle and fixed infrastructure, and between the vehicle and vehicles on adjacent tracks.

Certificate of Authority to Operate

Written notification by the infrastructure controller to a train operator of its **acceptance** of successful completion of the **Route Acceptance** process. This notification specifies the equipment, the equipment configuration, operational requirements and limitations, **route** constraints and network factors within which **acceptance** has been granted for network operations.

Certificate of Technical **Acceptance**

Written notification by the infrastructure controller to a train operator or third party, issued in cases where it is required to formalise the situation at an intermediate stage of the process leading to **Route Acceptance**. This notification confirms that a proposal specifying, where relevant, the equipment, the equipment configuration, operational requirements and limitations, **route** constraints and network factors, is acceptable at the date of issue as a full or partial basis for an application by a train operator for **route acceptance**.

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Certificate of Engineering Acceptance

Written notification by a vehicle acceptance body, issued as set out in GM/RT2000, that a vehicle meets the requirements of all applicable Railway Group Standards (except for any authorised non-compliances which are noted on the certificate - see GA/RT6001, GA/RT6004 and GA/RT6006) and which identifies any associated limitations on the operation of that vehicle.

Certificate of Gauging Acceptance

Written notification by the infrastructure controller that a vehicle has satisfactorily completed the gauging process for a route, undertaken by the infrastructure controller's designated person or persons.

Clearance

The calculated distance between vehicles and fixed structures or between two vehicles on adjacent tracks.

Comparative gauging

The process of comparing the swept envelopes of a vehicle new to a route, with the swept envelopes of a vehicle or vehicles which have been demonstrated to be able to use the proposed route.

Electromagnetic compatibility

Compatibility of electrical and electronic systems, in respect of emission and immunity levels, to ensure that operation of one system is not adversely affected to an unacceptable extent by emissions from any other system or equipment. For the purposes of this document, electromagnetic compatibility (EMC) assessment shall include consideration of conductive, inductive, capacitive and radiated effects at all frequencies from DC to 2 GHz.

Engineering acceptance

The process for confirming that a vehicle conforms to the mandatory requirements of the Railway Group Standards as set out in GM/RT2000.

Gauging

The process by which swept envelopes of a vehicle are used to determine clearances on a section of track between the vehicle and fixed structures and between the vehicle and vehicles on adjacent tracks.

Gauging acceptance

The endorsement by the infrastructure controller that a vehicle has satisfactorily completed the gauging process for a route.

Hybrid gauging

A combination of comparative and absolute gauging where absolute gauging is used to evaluate the clearances related to features of the vehicle projecting outside the envelope of the comparator vehicle.

Infrastructure systems

Fixed and portable systems and equipment forming part of the railway infrastructure provided or operated by the infrastructure controller. For the purposes of this document, infrastructure systems do not include:

- a) train-borne equipment (including train-borne components of signalling, control or communication systems, even where such systems are mandated or specified by the infrastructure controller)
- b) systems outside the boundary of the railway.

Interference

The potential of any characteristic or feature of a vehicle to impact adversely on the infrastructure at any interface with that infrastructure, or the potential of any characteristic or feature of the infrastructure to impact adversely on a vehicle at any interface with that vehicle.

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Lower sector

The area up to and including 1100 mm above the plane of the rails.

Notified Body

A registered Body appointed under the provisions of the Railways (High-Speed) (Interoperability) Regulations 2002 to confirm the compliance of railway systems with Technical Specifications for Interoperability (TSIs).

Passing clearance

The minimum calculated clearance at a location (at a position above the plane of the rails) between the swept envelopes of two specific types of rail vehicle as they pass on adjacent tracks at all speeds up to the maximum permissible line speeds.

Reduced clearance

A clearance, less than a normal clearance, which requires special measures to maintain tracks relative to adjacent tracks and structures.

Route

The physical path of a journey to be undertaken by a vehicle or a collection of vehicles, where the path comprises a number of sections of track each of which has individually defined characteristics.

Route acceptance

The process leading to **acceptance** by the infrastructure controller, and formalised by the issue of a Certificate of Authority to Operate, that all safety issues associated with the physical and operational characteristics of specified vehicles and their compatibility with the infrastructure on defined routes have been examined, and that the associated risks have been reduced to a level which is as low as reasonably practicable (ALARP).

Route Acceptance Safety Case

The document(s) which demonstrate the conditions under which vehicles can be operated safely on the defined **route**(s) and which forms the basis of an application for a Certificate of Authority to Operate. This is sometimes referred to as a 'System Safety Case'.

Route Availability

The assessed capacity of a **route** to carry the vertical static and dynamic loads of rail vehicles, expressed as a **Route Availability (RA) Number** as set out in [GE/RT8006](#).

The vertical static and dynamic loads of rail vehicles or the static load characteristic of a rail vehicle type, expressed as a **Route Availability (RA) Number** as set out in [GE/RT8006](#).

Section of track

Track bounded by identified limits such as junctions, terminals or points at which there is a significant change in traffic flow, or permissible speed.

Special reduced clearance

A clearance, less than a reduced clearance, which requires a specific risk assessment to be undertaken and the implementation of appropriate controls to demonstrate that risks have been reduced to as low as reasonably practicable (ALARP).

Standard vehicle gauge

An outline drawing or specification of a notional vehicle, which prescribes maximum permissible vehicle and loading dimensions, certain suspension displacements and certain curve overthrow limitations (for example, W6a).

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Susceptibility

The sensitivity of a vehicle to characteristics or features of the infrastructure at any interface with the infrastructure, or the sensitivity of the infrastructure to characteristics or features of the vehicle at any interface with that vehicle.

Swept envelope

A cross-sectional profile, taken at right angles to the track, enclosing all dynamic movements, static deflections and overthrows of all points along the surface of the vehicle, that can reasonably be expected to occur under the appropriate range of operating conditions as it sweeps past a theoretical track location. A family of swept envelopes is required to define a vehicle's behaviour on a route.

The swept envelopes referred to in this document exclude the effects of track tolerance and rail sidewear previously included in kinematic envelopes developed under GM/RT2149 issue 1 or earlier documents.

Technical Specification for Interoperability

Technical Specifications for Interoperability (TSIs) are mandatory standards forming part of the implementation of the European Interoperability Directives.

B4 Background

This section of the document provides a descriptive overview of the process and is included for guidance.

This document defines the process by which vehicles are accepted to run on Railtrack controlled infrastructure and by which the effects of changes to the infrastructure or vehicles are controlled.

It addresses the risks inherent in the operation of such vehicles on that infrastructure insofar as they have an effect on:

- a) the safe operation of the infrastructure
- b) safe interworking with other vehicles
- c) the safety of persons on or about the line
- d) the safety of the general public
- e) the safe operation of any adjacent or connecting railway systems.

It covers scheduled service operation, planned diversionary routes, movements to depots or workshops, testing and development, and any other occasional movements which may be required.

It does not address safety risks associated with the design of vehicle interiors where the responsibility for introducing the necessary risk controls lies entirely with the train operator and where the consequences of a failure of these controls is confined to persons within the vehicle. Nor does it address safety risks arising from activities of Railway Group members outside Railtrack controlled infrastructure other than where these can lead to risks being imported onto that infrastructure, through inadequate vehicle maintenance, for example.

The 'Railway (Safety Case) Regulations 2000 including 2001 amendments' (hereafter referred to as the Railway Safety Case Regulations) require the infrastructure controller to have an approved Railway Safety Case setting out those arrangements by which it will ensure a safe railway. A train operator is only permitted to operate vehicles on that infrastructure if it is the holder of an approved train operator's Railway Safety Case and if the proposed train operation has been demonstrated to be compatible with the infrastructure controller's approved Railway Safety Case.

Prior to demonstration of the compatibility of the vehicle with the infrastructure, the vehicle is subjected to Engineering Acceptance, the process whereby a

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Vehicle **Acceptance** Body confirms that the vehicle meets all mandatory requirements of Railway Group Standards applicable to that vehicle.

Route acceptance is the process whereby the infrastructure controller reviews the compatibility with its own Railway Safety Case of a train operator's proposed operational arrangements for a vehicle. The infrastructure controller also reviews with the train operator the compatibility of the rail vehicle and the **route** over which it is to operate. Upon satisfactory demonstration of compatibility the infrastructure controller authorises the operation by issuing a Certificate of Authority to Operate. The Certificate of Authority to Operate is the infrastructure controller's formal **acceptance** of an application from a train operator to run the vehicle under its Railway Safety Case on the specific routes identified, and identifies any operational limitations forming the conditions of that **acceptance**.

It is vital that the train operator and the infrastructure controller are involved in the submission for **route acceptance** at the earliest possible stage to ensure that operational implications as well as engineering aspects of vehicle design are taken fully into account. However, it is recognised that many train operators may wish much of the detailed work associated with the **acceptance** process to be done by a third party, and indeed may wish to purchase or lease vehicles which have already been demonstrated to satisfy certain or all of the conditions necessary for **acceptance**. This document therefore permits all or part of the work of establishing acceptable physical characteristics and associated operating requirements to be carried out, in conjunction with the infrastructure controller, by a suitably competent third party. Such third parties may wish to formalise the results of their work through an application to the infrastructure controller for a Certificate of Technical **Acceptance**, supported by appropriate evidence, which will ultimately form part of a **Route Acceptance** Safety Case. Such Certificates of Technical **Acceptance** only remain valid whilst the **route**, the vehicle and associated standards reviewed remain unchanged. Any subsequent changes in one or more of these aspects will require a new Certificate of Technical **Acceptance** or will require to be identified for consideration in the submission for a Certificate of Authority to Operate.

A Certificate of Authority to Operate may be sought in respect of a single vehicle, a group of vehicles of the same class or a fleet of vehicles of the same class.

With the introduction of The Railways (Interoperability) (High-Speed) Regulations 2002, vehicles coming within its scope have to comply with TSIs. For those vehicles coming under its scope Railway Group Standards only continue to be mandatory where the requirements are not fully addressed in the TSIs, and then only where the Railway Group Standard has been registered with the European Union as a notified standard. Vehicles coming within the scope of the above Regulations are approved by a Notified Body, rather than a Vehicle **Acceptance** Body. Whilst a body may be authorised to act as both a Notified Body and a Vehicle **Acceptance** Body, the roles are different in scope. A Vehicle **Acceptance** Body, in confirming conformity of a vehicle to Railway Group Standards, provides its certificate prior to **Route Acceptance**. A Notified Body, however, in confirming conformity of a vehicle to the TSIs, provides its Certificate of Conformity after **Route Acceptance** has been completed. The Railways (Interoperability) (High-Speed) Regulations 2002 has no recognition of **Route Acceptance** as a separate process outside the remit of the Notified Body. Whilst the network itself is not interoperable, the **Route Acceptance** process will remain. The **Route Acceptance** process requires that a Certificate of Engineering **Acceptance** is obtained prior to **Route Acceptance** being granted, whereas a Notified Body's Certificate of Conformity would not be issued prior to **Route Acceptance**. As an interim arrangement, therefore, Notified Bodies will issue a formal certificate confirming conformance as an input to the **Route Acceptance** process. This document therefore references the Notified Body process mandated by legislation and the interim arrangements, and describes the interfaces with the **Route Acceptance** process, but does not mandate the Notified Body process.

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Once the Regulations that implement the Conventional Interoperability Directive are published, the scope of the Notified Body process will be increased accordingly.

It should be noted that the Interoperability Regulations place responsibilities with the Contracting Entity (usually the vehicle purchaser, or owner) rather than the train operator. This **route acceptance** process requires that the applicant for the Certificate of Authority to Operate is the party introducing the vehicle to the infrastructure and therefore able to give all appropriate commitments to comply with any limitations attached to the Certificate of Authority to Operate.

It should also be noted that where vehicles come under the scope of 'The Railways (Interoperability) (High-Speed) Regulations 2002', or under the equivalent regulations for conventional vehicles, the Notified Body's Certificate of Conformity permits the Contracting Entity to issue a 'Declaration of Conformity' to the supervisory authority (the HSE in the U.K.). The vehicle can only enter service once authorised by the supervising authority. Vehicles not coming under the scope of the above regulations require approval from the HMRI in accordance with the 'Railways and Other Transport Systems (Approval of Works, Plant and Equipment) Regulations' as part of the authorisation of the proposed operation.

Appendix A gives a flow diagram illustrating the **route acceptance** process.

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Part C General requirements

C1 Management of route acceptance

The infrastructure controller shall manage and undertake route acceptance by means of a process conforming to this document.

The infrastructure controller shall establish a route acceptance panel, or panels, each comprising an identifiable body of people whose collective competencies are appropriate to a review of applications for route acceptance. The members of such panels shall be independent of the immediate project involved in an application, or projects competing with that for which the application has been made, to avoid conflicts of interest and to achieve an independent review of the proposal.

It is permissible for the panel or the train operator to appoint a third party as an Independent Safety Assessor (ISA) to validate the Route Acceptance Safety Case submitted and to provide advice on its acceptability. Where the train operator appoints an ISA, the ISA shall be subject to the agreement of the panel.

The route acceptance process shall include stages reflecting the needs of the applicant for route acceptance, including:

- a) stage 1 - concept or planning
- b) stage 2 - design
- c) stage 3 - development or testing
- d) stage 4 - implementation.

It is permitted to omit individual stages, or for all stages to be combined, by mutual agreement between the infrastructure controller and the train operator where appropriate.

C2 Circumstances requiring route acceptance

Route acceptance shall be obtained whenever:

- a) new vehicles are introduced onto Railtrack controlled infrastructure
- b) existing vehicles are required to operate on routes other than those over which they are currently authorised to operate
- c) the mechanical or electrical characteristics (including software) of vehicles, or the mode of operation of vehicles (including operational speed), are modified in a way which has a potential adverse material effect on safety at those operational interfaces forming the basis of route acceptance
- d) the infrastructure configuration is modified in a way which has a potential adverse material effect on the operating safety of any vehicles currently authorised to operate on a route, or those in the process of acceptance
- e) the infrastructure configuration is modified in a way which has a potential adverse material effect on its susceptibility to interference from any vehicles operating on it
- f) changes to the infrastructure improve the level of safety with the train operations current at that time, and the operator changes the vehicle or its operation to take advantage of the changes.

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C3 Applications for route acceptance

Applications for route acceptance originate from the train operator.

Route acceptance is the means of demonstrating that the operation of a vehicle along a route can be undertaken without prejudice to the infrastructure controller's approved Railway Safety Case. Therefore, no matter who originates the change requiring route acceptance, the train operator always originates the submission for route acceptance (see section D2.3).

Requests for route acceptance shall be submitted by the train operator to the infrastructure controller's route acceptance panel referred to in section C1 above.

Each request shall be accompanied by a Route Acceptance Safety Case describing the change requiring route acceptance, the risks introduced, and the mitigation measures proposed.

The Route Acceptance Safety Case shall demonstrate compatibility of new or modified vehicles with the infrastructure systems, or compatibility of the new or modified infrastructure with the vehicles operating over it.

C4 Management of gauging acceptance

The infrastructure controller shall manage and undertake gauging acceptance by means of a process conforming to this document.

The infrastructure controller shall identify and authorise a person or persons competent to undertake the process for granting gauging acceptance set out in section D2.8 and Appendix B of this document.

The authorised person or persons shall be competent to:

- identify the gauging criteria for the routes along which the vehicle is proposed to operate
- review the gauging portfolios produced as set out in the requirements of GM/RT2149 and submitted in support of an application for route acceptance
- identify any limitations on the acceptability.

The granting of gauging acceptance shall be confirmed by the issue of a certificate, signed by the authorised person. The certificate shall identify the vehicle or vehicles to which it applies, the route or routes for which it has been successfully gauged, and any operational limitations applicable.

C5 Electromagnetic compatibility

The infrastructure controller shall manage and undertake a review of the electromagnetic compatibility between the train and the infrastructure.

The infrastructure controller shall identify and authorise a person or persons competent to undertake the process for electromagnetic compatibility set out in section D2.9 and Appendix C of this document.

The authorised person or persons shall be competent to:

- identify the electromagnetic compatibility criteria for the routes along which the vehicle is proposed to operate
- review the declared vehicle electromagnetic interference and susceptibility data in relation to the criteria for the routes along which the vehicle is proposed to operate

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- c) identify any limitations on the acceptability.

The acceptability shall be confirmed by a declaration in writing, signed by the authorised person. The declaration shall identify the vehicle or vehicles to which it applies, the route or routes for which it has been successfully reviewed, and any operational limitations applicable.

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Part D

The route acceptance process

D1 The objective of route acceptance

The infrastructure controller's route acceptance panel shall satisfy itself that all safety issues associated with the characteristics of specified vehicles, or proposed grouping of vehicles, and their compatibility with the infrastructure on defined routes have been examined, and that the risks associated with the operation are not unacceptable and have been reduced to a level which is as low as reasonably practicable (ALARP).

The compatibility of a proposed train operation with the infrastructure controller's Railway Safety Case shall be assessed to confirm that:

- a) the vehicle is compatible with, and can safely interwork with, the infrastructure and other users of that infrastructure
- b) the controls on the proposed operation are compatible with the infrastructure controller's Railway Safety Case.

D2 The process leading to issue of a Certificate of Authority to Operate

D2.1 Exceptions to the process

In seeking a Certificate of Authority to Operate, the train operator shall follow the process set out in this section, and summarised in Appendix A, except as set out below:

- a) Evidence submitted for similar vehicles having identical characteristics on the same route or routes also having identical characteristics shall be accepted.
- b) In the case of freight vehicles built to a standard vehicle gauge applicable to the route concerned, the application of the process defined below shall be limited only to those interface issues specific to the vehicle concerned.
- c) Where the application is as a result of a change to the infrastructure, and does not require any change to the vehicle, D2.7, D2.8 and D2.11 do not apply.

D2.2 Exchange of information

There shall be an exchange of information between the train operator and the infrastructure controller in the case of vehicle proposals, and between the infrastructure controller and all the potentially affected train operators in the case of infrastructure proposals. In each case, this exchange shall identify, but not be limited to, the likely consequences on safety, potential mitigation measures and additional information requirements. The information exchanged shall include, but is not limited to, that referred to in sections G2 and H2 of this document.

D2.3 Identification of risks

The party proposing the action requiring route acceptance shall lead a risk assessment, to identify the risks associated with the proposals (including the operational risks), in conjunction with the other affected parties. There shall then be an analysis to identify which risks are addressed by compliance with the applicable mandatory requirements (see sections G1 and H1) and which are not. In the latter case, the parties shall determine a strategy for reducing risks falling within the scope of this document to a level which is not intolerable and is as low as reasonably practicable (ALARP).

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D2.4 Acceptance criteria

Following determination of the strategy for reduction of risk, the infrastructure controller and the train operator shall submit the parameters and criteria identified as set out in Part F of this document to the **route acceptance** panel for endorsement.

The **route acceptance** panel shall give clear confirmation of the criteria by which it will judge the information presented in support of an application for a Certificate of Authority to Operate, and shall indicate the nature and scope of the evidence which it requires to be presented.

D2.5 Division of responsibility

When the overall risk control strategy has been identified and endorsed, the train operator and the infrastructure controller shall, led by the party proposing the action requiring **route acceptance**, produce a clear documented statement defining who is responsible for the implementation of the necessary work and the timescales, and who is responsible for foreseeable consequences of the proposed change that could emerge subsequently. The statement shall identify a clear link between the assumptions made in the evidence supporting the application for **acceptance** and the actual arrangements which apply.

D2.6 Acceptability of the proposals

The **route acceptance** panel shall apply the criteria identified as set out in Part F of this document in assessing the information presented, both in respect of what constitutes acceptable performance and in respect of the integrity of the process by which that performance is assessed.

The **route acceptance** panel shall indicate clearly to the train operator whether the evidence developed as set out in sections D2.2 to D2.4 above, and presented in support of the application for a Certificate of Authority to Operate, represents an acceptable demonstration that the risk has been reduced to a level which is not intolerable and which is as low as reasonably practicable (ALARP). If the evidence is not endorsed, the **route acceptance** panel shall give a clear statement of the reasons why, and shall indicate what further steps need to be taken by the train operator or the infrastructure controller (in accordance with the division of responsibility agreed in section D2.5 above) in order to provide such a demonstration.

D2.7 Engineering acceptance

Prior to issue of a Certificate of Authority to Operate, the train operator shall obtain a Certificate of Engineering **Acceptance**, except as indicated in the following paragraph, for the vehicles concerned. Any limitations on operation with which the latter certificate is endorsed, and which affect **route acceptance**, shall be taken into account by the infrastructure controller in determining the conditions under which a Certificate of Authority to Operate is issued.

In the case of vehicles covered by The Railways (Interoperability) (High-Speed) Regulations 2002, or those covered by the equivalent regulations for conventional vehicles, written confirmation that the vehicle conforms with applicable mandatory requirements shall be obtained by the train operator from the Contracting Entity, who in turn obtains it from the appointed Notified Body reviewing compliance of the vehicles.

D2.8 Gauging acceptance

Prior to issue of a Certificate of Authority to Operate, the train operator shall obtain a Certificate of Gauging **Acceptance** for the vehicles concerned from the infrastructure controller (see section C4). Any limitations on operation with which the Certificate of Gauging **Acceptance** is endorsed shall be taken into account by the infrastructure controller in determining the conditions under which a Certificate of Authority to Operate is issued.

Where vehicles have systems installed that enable the body of the vehicle to tilt, the clearances throughout the **route** shall be assessed for the vehicle in normal operation and failure modes as required by [GE/RT8019](#). The review shall also

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include the compatibility between the vehicle and **route** where clearances along the **route** require a Tilt Enable and Supervision System.

Applications for a Certificate of Gauging **Acceptance** shall be based on one of the gauging procedures set out in Appendix B. The infrastructure controller shall determine which method is appropriate to the definition of the **route** for which **route acceptance** is sought.

D2.9 Confirmation of electromagnetic compatibility

Prior to issue of a Certificate of Authority to Operate, the train operator shall also obtain a declaration in writing confirming the electromagnetic compatibility of the vehicles concerned with the proposed routes, from the infrastructure controller (see section C5). Any limitations on operation identified in the declaration shall be taken into account by the infrastructure controller in determining the conditions under which a Certificate of Authority to Operate is issued. (See also Appendix C).

D2.10 Issue of a Certificate of Authority to Operate

Prior to issuing a Certificate of Authority to Operate (including a temporary Certificate of Authority to Operate as set out in section D2.11), the **route acceptance** panel shall be satisfied that:

- the process set out in sections D2.2 to D2.9 has been followed
- the evidence presented by the train operator and the infrastructure controller represents an adequate demonstration that the risks have been reduced to a level which is not intolerable and is as low as is reasonably practicable
- there are robust arrangements in place for the timely implementation of the actions necessary to achieve the agreed risk controls, in cases where certification is conditional on changes.

D2.11 On-track testing

Where, as part of the process, on-track testing needs to be carried out on Railtrack controlled infrastructure, the testing shall take place under a train operator's Railway Safety Case, whether that train operator is engaged only for the purpose of the testing or is the ultimate operator of the vehicles. A Certificate of Engineering **Acceptance**, or confirmation of conformance from a Notified Body (see section D2.7), and a temporary Certificate of Authority to Operate for this specific purpose shall be obtained. This latter certificate shall identify all required additional risk control measures and special operating procedures. It shall be time limited to cover in most cases the required test period only.

D3 Involvement of third parties in the **route acceptance** process

Whilst a Certificate of Authority to Operate is issued to a train operator only, it is permissible for a competent third party (such as a manufacturer or a train leasing company or a specialist consultant) to carry out some or all of the tasks in pursuit of a Certificate of Authority to Operate which are specifically assigned to the train operator in section D2.

Where there is a need to formalise what has been accepted, for example as part of product development and where the end-user train operator has not been identified, this shall be by means of a Certificate of Technical **Acceptance**. Where this is issued on a partial basis (that is where it does not represent an **acceptance** that all the necessary conditions for reducing the risk to a level as low as reasonably practicable (ALARP) have been met), the certificate shall be endorsed to indicate its scope, and to indicate the issues which remain.

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Where a train operator seeks a Certificate of Authority to Operate which involves reference to a Certificate of Technical **Acceptance** obtained by a third party, the train operator shall:

- a) satisfy itself that the physical characteristics, performance and mode of operation of the vehicles intended to be operated on Railtrack controlled infrastructure are as set out in the evidence put forward in support of the application for the Certificate of Technical **Acceptance**
- b) ensure that any outstanding issues are dealt with by the processes set out in section D2.

D4 Interim **acceptance**

In those circumstances where it is not feasible or desirable for the full **route acceptance** process to be completed immediately for the intended service operation envisaged, the train operator and the infrastructure controller shall determine a phased programme of **acceptance**. It is permissible for a third party to act in advance of a train operator. An example of such circumstances is where either the vehicles or the infrastructure or both are not in their final state, or because the adequacy of the proposed arrangements needs to be demonstrated through on-track testing or a period of service evaluation.

In a situation where the programme involves on-track running, the infrastructure controller and the train operator shall ensure that the processes set out in section D2 have been carried out and outputs documented. In particular they shall ensure that any risks associated with the unproven nature of the vehicles or the infrastructure or of their mutual interaction have been addressed by the introduction of appropriate temporary control measures. This shall include consideration of any endorsements on the Certificate of Engineering **Acceptance** or Notified Body confirmation of conformance, which may itself be provisional, awaiting the results of track tests or satisfactory service experience.

D5 Disputes

D5.1 Right of appeal

Where there is disagreement between the infrastructure controller and the train operator/third party, and the safety of the operation is considered by either party to be in question, there shall be a right of appeal to Railway Safety. Such areas of disagreement include, but are not limited to, the following:

- a) the appropriateness of the models, algorithms and analysis methods used in any assessment of performance
- b) the appropriateness of the data used as the basis for such an assessment
- c) the acceptability of the final result
- d) the criteria for **acceptance**
- e) the quality and completeness of the assessment process.

D5.2 Resolution

When an appeal is made, Railway Safety shall appoint in a timely manner appropriate specialist advisors, acceptable to both parties, to make recommendations as to how the intended operation can be safeguarded.

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Part E Management of change

E1 Change to the infrastructure

E1.1 Changes affecting the susceptibility of vehicles to interference from the infrastructure

The infrastructure controller shall review the effects of any planned changes to the infrastructure, including construction or modification of a station, where there could be an adverse effect on the safe operation of vehicles. The infrastructure controller shall involve all the train operators or third parties (for example, those in possession of a Certificate of Technical **Acceptance**) who are potentially affected and shall provide, within a mutually agreed timescale, any additional infrastructure data requested which are needed to permit the effects of the change to be assessed.

E1.2 Changes affecting the susceptibility of the infrastructure to interference from vehicles

The infrastructure controller shall review the effects of any planned changes to the infrastructure that would make the infrastructure or its systems more susceptible to interference from vehicles in any interface aspect. The requirement for the review includes the case where the proposed change interferes with a standard vehicle gauge for which the **route** has previously been cleared thus reducing **route** capability. The review shall consider the effect of this change on the validity of the **route acceptance** for all vehicles authorised to operate over the **route** (including those vehicles authorised to operate by processes which preceded the issue of Railway Group Standards). Train operators shall provide, within a mutually agreed timescale, any additional vehicle data reasonably requested by the infrastructure controller (in accordance with the procedures set out in this document) which are needed to permit the effects of the change to be assessed.

E1.3 Review of the safety performance of the system

Even though the initiative for a change comes from the infrastructure controller, the process requires that each Certificate of Technical **Acceptance** and each Certificate of Authority to Operate take account of all the relevant risks. If those risks subsequently change, the continued validity of the certificates shall be reviewed. The end result of a proposed infrastructure change could be:

- a) A conclusion that no action is necessary in respect of the physical characteristics of vehicles or their mode of operation, in which case the existing Certificate of Authority to Operate continues to be valid.
- b) A requirement to alter either the physical design or mode of operation of vehicles or infrastructure, or both, in which case it would be necessary to examine all the safety implications of the changes to both vehicles and infrastructure and obtain a Certificate(s) of Authority to Operate on that revised basis. Changes to the vehicle will also require approval under the Engineering **Acceptance** process or by a Notified Body as appropriate.

The infrastructure controller shall therefore support the train operator's demonstration with a **Route Acceptance** Safety Case that the safety performance of the system will not be reduced by operation of its existing train formations. This shall take into account normal, degraded and fault conditions on the vehicle and the infrastructure.

E2 Change to the vehicle or its usage

E2.1 Changes to an interface reviewed under **route acceptance**

In cases where there is a proposal to change the vehicle, and where such change affects an interface reviewed under **route acceptance**, the train operator shall submit full details to the **route acceptance** panel referred to in section C1

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for a review of the vehicle's **route acceptance**. Change to the vehicle having no effect on the interfaces, even if requiring Engineering **Acceptance**, shall not require resubmission for **route acceptance**.

E2.2 Changes to train formation or to the use of an existing formation

In cases where there is a proposal to modify a train formation, or change the use of an existing train formation, the train operator shall submit full details to the **route acceptance** panel referred to in section C1 for a review of the vehicle's **route acceptance**. The submission shall consider the potential effects throughout the period that the train formation is intended to operate on the **route**. It shall take into account normal, degraded and fault conditions on the vehicle and of the infrastructure.

E3 Change to the authorised operation of vehicles affecting stations

E3.1

The train operator shall consult with the infrastructure controller and the relevant station operator before the operation of railway vehicles passing or stopping at a station is changed in the following situations:

- a) where the maximum speed of a railway vehicle as it passes the station is increased
- b) where passenger vehicles new to the **route** are booked to stop at the station
- c) where passenger vehicles operating on the **route** are newly booked to stop at the station
- d) where the train formation or use of a train formation is changed as indicated in section E2.2 above.

E3.2

The station operator shall review the impact on its operation of the change to the authorised train operation and identify those safety issues requiring mitigation prior to authorisation of the operational change.

E3.3

The train operator shall include within its **Route Acceptance** Safety Case those issues raised by the station operator and the mitigation measures by which the issues are to be addressed.

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Part F

Acceptance parameters and criteria

F1 Parameter definition

The infrastructure controller and the train operator shall jointly identify the interfaces between the vehicle and the infrastructure relevant to the **route** or routes along which the vehicle is proposed to operate.

The infrastructure controller and the train operator shall jointly define the parameters for reviewing the compatibility of the vehicle and infrastructure at the interfaces. These parameters shall include, though not be limited to, the following:

- a) those defining the physical characteristics of the infrastructure and the vehicle, including gauging, stepping distance and **route** availability (RA) number
- b) those defining the electrical characteristics of the infrastructure and the vehicle, including the infrastructure's supply and return systems, capacities and location; and the electrical power demands of the vehicle, its duty cycles and characteristics, and its current collection systems (where appropriate)
- c) those defining the electromagnetic characteristics of the infrastructure and the vehicle and, where practicable, the characteristics of neighbours or neighbouring railways
- d) those defining the train control and communications system
- e) those defining the dynamic interaction between the infrastructure and the vehicle, including abnormal track forces and the effects of bridge resonance
- f) those defining the functions and characteristics of the operational interface systems between the infrastructure and the vehicle (for example, door operating systems and controls).

F2 Identifying the criteria

Using a risk-based approach, the infrastructure controller and train operator shall identify **acceptance** criteria for each of the interface parameters identified under section F1 above, by taking into consideration the following:

- a) the potential of the vehicle to interfere with the **route**, its other users or its neighbours
- b) the potential for the **route**, its other users or its neighbours to interfere with the vehicle
- c) the susceptibility of the **route**, its other users or its neighbours to the characteristics of the vehicle
- d) the susceptibility of the vehicle to the characteristics of the **route**, its other users or its neighbours.

The basis of the determination shall be submitted as part of the **Route Acceptance** Safety Case.

The risk-based approach to identifying criteria of **acceptance** for each of the parameters shall identify the risks arising from the change proposed to the infrastructure (by its construction, modification or re-definition) or the vehicle (by its introduction, revised operation or modification). A reduction in risk with respect to one or more of the criteria relating to a parameter shall not permit an

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increase in risk with respect to other criteria relating to that or any of the other parameters.

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Part G Vehicle status and performance

G1 Evidence of compatibility

As part of the submission for **route acceptance**, the train operator shall:

- a) provide confirmation that the vehicle itself meets applicable mandatory standards
- b) confirm, where the vehicle remains unchanged, and does not meet mandatory standards, the current status of the vehicle
- c) provide validated interface data with respect to a) and b) above by which the vehicle's compatibility with the infrastructure can be assessed.

The demonstration that the vehicle meets all applicable mandatory requirements is either by means of a Certificate of Engineering **Acceptance** or by a specific confirmation from a Notified Body as indicated below. Where the vehicle remains unchanged, the current Certificate of Conformance is required.

In the case of modification to vehicles subject to The Railways (Interoperability) (High-Speed) Regulations, compliance with Technical Standards for Interoperability (TSIs) and, where applicable, those Railway Group Standards that have become Notified Standards, shall be confirmed by an appointed Notified Body.

In the case of modification to vehicles not subject to the above regulations, compliance with all relevant Railway Group Standards shall be confirmed by a Certificate of Engineering **Acceptance** issued by an authorised Vehicle **Acceptance** Body.

The process relating to high-speed interoperability, referred to above, will equally apply to conventional vehicles when the equivalent regulations are introduced.

The confirmation that the vehicle conforms with the applicable mandatory requirements, either by the Notified Body's written confirmation or the Certificate of Engineering **Acceptance**, as applicable, is required before a Certificate of Authority to Operate can be granted. It is permissible to proceed in parallel with the preparatory work required to obtain both the confirmation and the Certificate of Authority to Operate.

In cases where vehicle related information to be presented to the infrastructure controller is mandated by Railway Group Standards, and where that information is an essential part of the **route acceptance** process, the integrity of the process by which the information is generated shall be demonstrated through the Certificate of Engineering **Acceptance**. This indicates that the signatory of that certificate is satisfied that the information presented to the infrastructure controller has been obtained by competent persons in an approved manner. An example of the above is the need to produce swept envelopes as part of the process of ensuring that vehicles satisfy gauging requirements.

G2 Interface data

G2.1 General

The train operator shall create and maintain the data defining the status and performance of its vehicles, that data shall include, but not be limited to, that set out in sections G2.2 to G2.5.

G2.2 Gauging

The train operator shall provide all vehicle gauging data, defined and validated as set out in [GM/RT2149](#) relevant to the **route acceptance** submission. This data shall include a clear definition of the pantograph size, interface

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characteristics and sway movements. (See also section D2.8 and Appendix B of this document).

Where the vehicle size is defined by swept envelopes, the swept envelopes for the conditions applicable to the proposed route shall be selected. The swept envelopes for the wind speeds applicable to the route to be assessed shall be selected. Where vehicles have systems installed that enable the body of the vehicle to tilt, swept envelopes shall be provided for the vehicles under normal tilt and credible failure modes as required by GM/RT2149 and GE/RT8019.

Where the train operator proposes to operate vehicles having systems that enable the body of the vehicle to tilt, it shall submit a full description of the system operation, controls and the performance characteristics at the vehicle/infrastructure interface. Tilt Enable and Supervision Systems shall be described, including the response characteristics to tilt prohibition signals, and complementary requirements for infrastructure based equipment shall be identified as required by and defined in GE/RT8019. The train operator shall also declare any provisions in place for identifying the permissible speed to the driver where the vehicle is not permitted to operate at the enhanced permissible speeds applicable to the route, as set out in GE/RT8012. A risk assessment of the operation shall be included within the risk assessment required by section D2.3 of this document.

G2.3 Electromagnetic compatibility

The train operator shall identify those vehicle systems potentially generating electrical and electromagnetic interference, as set out in GE/RT8015, and shall provide data on the anticipated emission levels. This shall take into account normal, degraded and foreseeable fault conditions on the vehicle. (See also Appendix C of this document).

The train operator shall also identify those vehicle systems susceptible to such interference, as set out in GE/RT8015, and those thresholds above which the vehicle systems could be susceptible. This shall take into account normal, degraded and foreseeable fault conditions on the vehicle. (See also Appendix C of this document).

G2.4 Route availability

The train operator shall, as set out in GE/RT8006, declare in the route acceptance submission the RA Number, the maximum vehicle speed, the axle spacings, vehicle overhang, maximum static tare and gross laden axle loads, and the maximum height of the centre of gravity in the laden condition.

G2.5 Compatibility with electrification system

The train operator shall, as set out in GE/RT8023, review the compatibility with the electrification system of any electrically powered vehicle new to the route, or of any change to an existing electrically powered vehicle, or of any change to the operation (for example, in the timetable, train configuration or the operational speed) of the vehicle.

G2.6 Wheel-rail interface

G2.6.1

Specific submissions for route acceptance shall not be required where the vehicle is fully compliant with GM/TT0088, and that compliance is certificated by Engineering Acceptance. Where the vehicle is not fully compliant with the requirements of GM/TT0088, the requirements of G2.6.2 and G2.6.3 shall apply.

G2.6.2

The train operator shall, as set out in GM/TT0088, declare proposals to operate vehicles with combinations of static wheel loading, wheel diameter and wheel tread profile that are not in accordance with GM/TT0088, together with evidence that the wheel-to-rail contact stresses and deformation rates are maintained within safe limits.

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G2.6.3

The train operator shall, as set out in [GM/TT0088](#), declare proposals to operate vehicles which exceed the maximum values of the vertical dynamic (P2) force, the lateral (Y) force and the longitudinal forces permitted by that standard for specific approval in relation to the routes for which **route acceptance** is sought.

G3 Configuration control of vehicles

The final responsibility for the performance of the vehicle rests with the train operator. The train operator, in seeking a Certificate of Authority to Operate, shall therefore confirm that the vehicle for which **acceptance** is sought is as described in the evidence put forward in support of the application.

The train operator shall have control over the configuration of vehicles such that they remain materially in the 'as accepted' condition or that systems are in place to assess the impact of any changes which could trigger a need for resubmission.

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Part H Infrastructure status and performance

H1 Evidence of compatibility

In order to facilitate a submission for **route acceptance** from a train operator, or as a basis on which to assess the compatibility between a vehicle and the **route** over which the vehicle is to operate, the infrastructure controller shall:

- a) confirm whether the infrastructure meets mandatory standards appropriate to the **route acceptance** submission
- b) confirm, where the infrastructure remains unchanged, and does not meet mandatory standards, the current configuration of the infrastructure
- c) provide validated interface data with respect to a) and b) above by which the vehicle's compatibility with the infrastructure can be assessed.

The demonstration that the infrastructure meets all mandatory requirements relevant to the **route acceptance** submission will either be by means of specific confirmation of status and the provision of mandatory data or by a Certificate of Conformity from a Notified Body (as indicated below).

In the case of upgrades to routes subject to The Railways (Interoperability) (High-Speed) Regulations, compliance with TSIs and, where applicable, those Railway Group Standards that have become Notified Standards, shall be confirmed by an appointed Notified Body. In the case of routes not subject to these regulations, the infrastructure controller shall implement a process for confirming compliance with Railway Group Standards.

H2 Interface data

H2.1 General

The infrastructure controller shall create and maintain the data defining the status and performance of its infrastructure. That data shall include, but not be limited to, that set out in sections H2.2 to H2.4.

H2.2 Gauging

The physical characteristics of the infrastructure shall be defined and validated as set out in [GC/RT5212](#) and the infrastructure controller shall submit the relevant data to the train operator as the basis for reviewing the compatibility of the vehicles to operate on that **route**.

The conditions applicable to the proposed **route** shall be identified to permit the applicable vehicle data (for example, the swept envelopes) to be provided by the train operator.

For routes where the operation of tilting vehicles is proposed, the infrastructure controller shall identify those sections of the **route** where tilting operation is permissible and identify tilt prohibited sections where tilting operation is not permitted, as required by and defined in [GE/RT8019](#). The infrastructure controller shall identify the infrastructure elements of the Tilt Enable and Supervision System, and their performance characteristics at the vehicle/infrastructure interface. The infrastructure controller shall also declare the enhanced permissible speed limits for the sections of the **route** over which tilt operation is permitted, and any provisions in place for tilting trains that are not permitted to operate at the enhanced permissible speeds applicable to the **route**, as set out in [GE/RT8012](#).

H2.3 Electromagnetic compatibility

The infrastructure controller shall identify those infrastructure systems potentially generating electrical and electromagnetic interference, as set out in [GE/RT8015](#), and shall provide data on the anticipated emission levels.

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The infrastructure controller shall also identify those infrastructure systems susceptible to such interference, as set out in [GE/RT8015](#), and those thresholds above which the infrastructure systems could be susceptible. This shall take into account normal, degraded and foreseeable fault conditions on the infrastructure.

H2.4 Route availability

The infrastructure controller shall determine the **Route** Availability (RA) Number at the permissible speed for the **route**, and at intermediate speeds where necessitated by a vehicle with higher RA Number, in accordance with the principles and methodology set out in [GE/RT8006](#).

H2.5 Compatibility of the electrification system with the vehicles using it

The infrastructure controller shall, as set out in [GE/RT8023](#), review the compatibility of any change to the electrification system with the vehicles using that system. Electrical clearances shall be confirmed as compliant with [GE/RT8025](#).

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Part I

Responsibilities of the infrastructure controller

I1 Organisation and competence

The infrastructure controller shall have processes and resources to fulfil its obligations as set out in this document in a timely manner. In particular the infrastructure controller shall:

- a) have clearly defined arrangements for interfacing with train operators or third parties for the purposes of **route acceptance**
- b) operate within a defined framework which ensures that all safety aspects of vehicle **acceptance**, including granting of gauging **acceptance**, are dealt with in a systematic and controlled manner
- c) ensure that the people with appropriate competences are made available to fulfil its obligations under this document
- d) ensure that sufficient information is provided regarding the physical characteristics and behaviour of the infrastructure, in order for the processes required by this document to be satisfactorily completed
- e) ensure the accuracy of any information regarding the physical characteristics and behaviour of the infrastructure which it supplies for use in the processes required by this document
- f) have processes in place to control the configuration of infrastructure such that, unless the processes set out in this document have been invoked, no changes are made which have the potential to adversely affect the compatibility with vehicles authorised to operate over it, or for which a Certificate of Technical **Acceptance** has been granted
- g) ensure that those responsible for the planning and implementation of infrastructure modifications and renewals are aware of the potential implications on safe operation of vehicles; that they also have available sufficient information to allow them to judge whether a proposed change requires **route acceptance** to be invoked or whether it is necessary to consult with train operators in order to decide whether it needs to be invoked
- h) ensure that there are processes to advise holders of Certificates of Technical **Acceptance** and Certificates of Authority to Operate in a situation where infrastructure change potentially affects the validity of these documents.

I2 Record keeping

I2.1 Informing relevant train operators of details of granted Certificates of Authority to Operate

The infrastructure controller shall have a defined process for informing relevant train operators of details of Certificates of Authority to Operate which have been granted, stating specifically:

- a) the vehicles involved
- b) the exact routes over which **acceptance** is granted
- c) any conditions or restrictions which are applicable
- d) the date of authorisation.

I2.2 Recording details of granted Certificates of Authority to Operate

The infrastructure controller shall have a defined process for recording the information in section I2.1 above, in respect of authorisation granted via the

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process set out in this document, or by the process previously used by British Rail, in order that the infrastructure controller is able to:

- a) fulfil its obligations in respect of the processes set out in this document
- b) ensure that day-to-day operations are conducted in accordance with the authorisations which have been granted and that any associated limitations are adhered to
- c) ensure that there is a record of vehicles authorised to operate on a particular **route** in cases where actual operation has ceased, so that the implications of proposed infrastructure changes on any such vehicles can be assessed without unnecessary repetition of the processes set out in this document.

There shall be a clear definition of the configuration of the vehicle for which the Certificate of Authority to Operate has been granted. (See also [GM/RT2453](#) and [GE/RT8006](#)).

The infrastructure controller shall retain auditable records of the evidence upon which **route acceptance** has been granted.

12.3 Records of vehicles having gauging **acceptance** on each section of track

For each section of track the infrastructure controller shall record details of vehicles permitted to operate on that section and the speeds at which they are permitted to run.

The permitted vehicles shall be classified as follows:

- a) those which have been gauged in accordance with the requirements of this document and [GM/RT2149](#) (issue 2 or later)
- b) those which have not been gauged in accordance with the requirements of this document and [GM/RT2149](#) (issue 2 or later) - for example, vehicles for which limited gauging has been undertaken or which have been previously gauged using a system other than that set out in Appendix B, sections B1, B2, B3 and B4.

Specific restrictions at individual locations shall be recorded (for example, a speed restriction passing a particular structure).

The infrastructure controller shall keep a record of the vehicle gauging portfolio (as set out in [GM/RT2149](#)) for each vehicle reflecting its status at the time of obtaining gauging **acceptance**. These records shall be kept for a minimum period of three years after cessation of the vehicle's operation on a particular **route**.

12.4 Records of vehicles verified as having electromagnetic compatibility for each section of track

For each section of track the infrastructure controller shall record details of vehicles for which electromagnetic compatibility has been verified and any specific restrictions applying.

The infrastructure controller shall record the date at which the vehicle obtained electromagnetic compatibility verification.

These records shall be kept for a minimum period of three years after cessation of the vehicle's operation on a particular **route**.

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Part J Responsibilities of train operators

J1 Organisation and competence

Train operators shall have processes and resources to fulfil their obligations set out in this document in a timely manner. In particular train operators shall:

- a) have clearly defined arrangements for interfacing with the infrastructure controller for the purposes of route acceptance
- b) operate within a defined framework which ensures that all safety aspects of vehicle acceptance are dealt with in a systematic and controlled manner
- c) ensure that people with the appropriate competences are made available to fulfil its obligations under this document
- d) ensure that sufficient information is provided regarding the physical characteristics and behaviour of the rolling stock, in order for the processes required by this document to be satisfactorily completed
- e) ensure the accuracy of any information regarding the physical characteristics and behaviour of the rolling stock which it supplies for use in the processes required by this document
- f) have processes in place to control the configuration of vehicles such that, unless the processes set out in this document have been invoked, no changes are made which have the potential to affect their compatibility with the infrastructure.

J2 Record keeping

Train operators shall:

- a) have a defined process for recording details of Certificates of Authority to Operate which have been granted
- b) ensure that there are auditable records of the work which has been done to fulfil its obligations under this document
- c) keep the records for a minimum period of three years after cessation of the vehicle's operation on a particular route.

J3 Control of operations

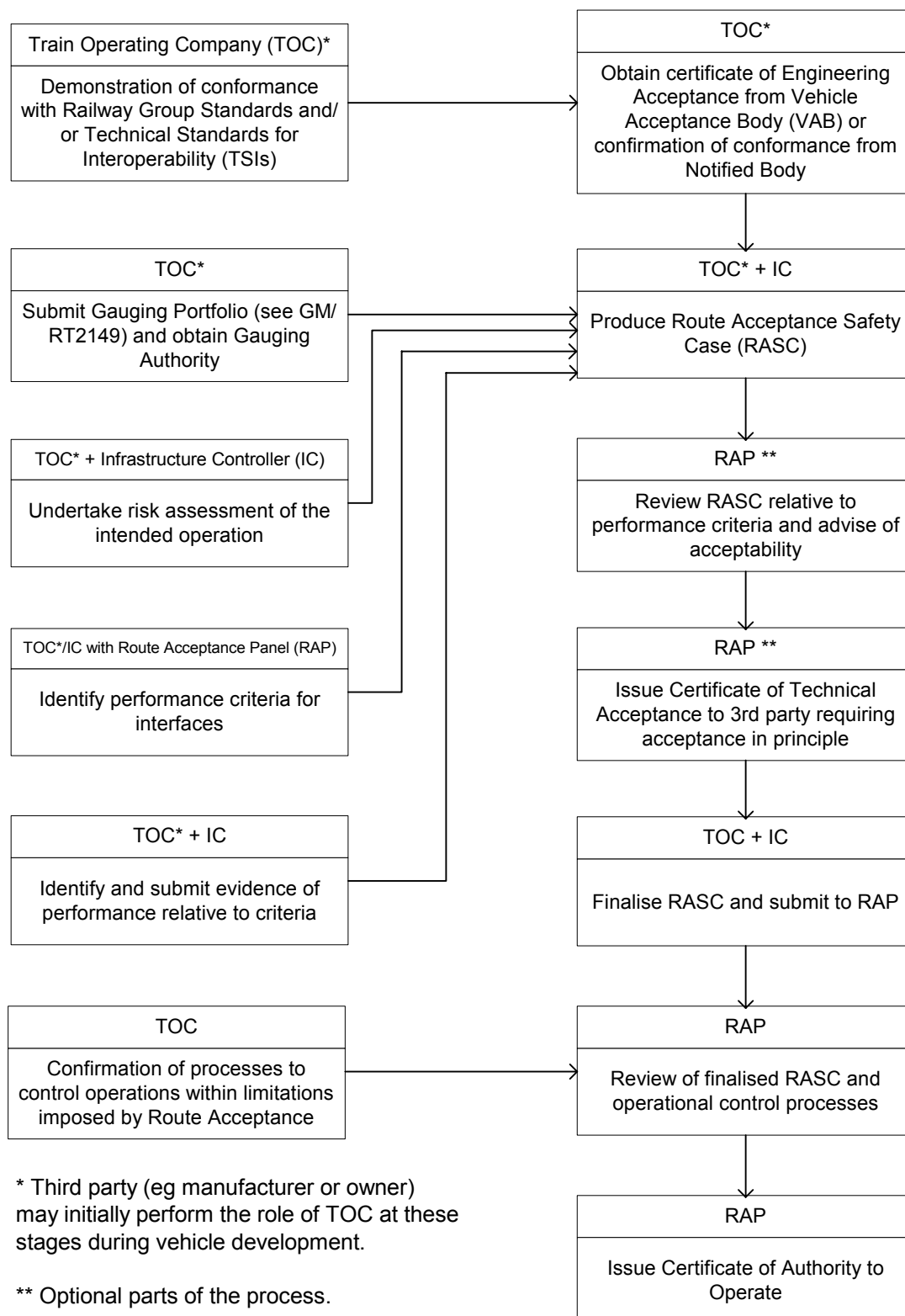
J3.1 Train operator's compliance with Certificates of Authority to Operate granted

The train operator shall have a documented process by which it ensures that all vehicles operated under its Railway Safety Case:

- a) have been authorised via the process set out in this document or by the processes which preceded its issue
- b) are confined to the routes for which authority has been granted
- c) are operated in accordance with any conditions or restrictions relevant to that authority.

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Appendix A Route Acceptance Flow Diagram



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Appendix B Procedures for gauging acceptance

(The content of this appendix is mandatory).

This appendix sets out the gauging procedures to be followed when applying for a Certificate of Gauging **Acceptance** as set out in section D2.8.

B1 Absolute gauging

Absolute gauging shall be undertaken unless one of the alternative methods set out in sections B2 to B4 has been agreed between the train operator and the infrastructure controller.

B2 Gauging to standard vehicle gauges

Vehicles conforming to a standard vehicle gauge shall not require specific gauging **acceptance** where the **route** concerned has been published by the infrastructure controller as accepting that particular standard vehicle gauge.

B3 Comparative gauging: comparison with vehicles already cleared for the **route**

Where comparative gauging using swept envelopes is to be used, the swept envelopes shall have been produced using compatible methodologies as set out in the requirements of [GM/RT2149](#) (issue 2 or later).

Vehicles chosen for comparison shall be those that comprise significant and regular traffic on the **route(s)** being considered.

If all swept envelopes of a candidate vehicle are within the swept envelopes of a comparator vehicle, or vehicles, at present using the section of track, the request for gauging **acceptance** shall be based on the authority for the comparator vehicle.

Any reduced or special reduced clearances agreed for the comparator vehicle shall be transferred to the candidate vehicle, subject to confirmation of the risk assessment for special reduced clearances required by [GC/RT5212](#).

B4 Hybrid gauging

Where parts of swept envelopes of the candidate vehicle are outside the swept envelopes of a comparator vehicle using the section of track, then either:

- a) those parts of the swept envelope exceeding the swept envelope of the comparator vehicle shall be demonstrated to be within the swept envelope of another class of vehicle accepted by the infrastructure controller as a valid comparator vehicle for the **route**, or
- b) absolute gauging shall be carried out on those parts of the vehicle for the specific features of each section of track causing the swept envelopes of the vehicle to exceed those of the comparator vehicle.

Any reduced or special reduced clearances agreed for the comparator vehicle shall be transferred to the candidate vehicle, subject to confirmation of the risk assessment for special reduced clearances required by [GC/RT5212](#).

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B5 Size of vehicles within the lower sector

For proposed introductions of vehicles new to a **route** an assessment shall be carried out to determine whether there are any restrictions due to:

- a) the vehicle's size at positions within the lower sector not complying with requirements for new vehicles (as set out in [GM/RT2149](#) issue 2 or later)
- b) infrastructure on intended routes does not yet conform to the structure gauge as set out in [GC/RT5212](#)
- c) specific locations not conforming to the gauge but with derogations in place.

B6 Gauging **acceptance** for existing vehicles on existing routes

Vehicles already operating on specific routes prior to the date this document comes into force shall be deemed to have gauging **acceptance** for those specific routes.

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Appendix C Demonstration of electromagnetic compatibility

(The content of this appendix is mandatory).

This appendix sets out the demonstration of electromagnetic compatibility required of the train operator or infrastructure controller in the event of a change.

C1 Vehicles

C1.1

Where a new vehicle, a modified vehicle, or a vehicle new to the **route**, is to be introduced, or where a change to operations is proposed, the train operator shall demonstrate its compatibility with the infrastructure systems as set out in sections C1.2 to C1.4 below.

C1.2

The train operator shall demonstrate compatibility of modified vehicles, or vehicles new to the **route**, with the infrastructure systems, using the information provided by the infrastructure controller, and the methodology for demonstrating that compatibility as set out in Railway Group Standards [GE/RT8015](#).

C1.3

The train operator shall demonstrate that the safety performance of train-borne systems will not be compromised by electromagnetic interference from the infrastructure at the levels declared by the infrastructure controller.

C1.4

The train operator shall, using the infrastructure susceptibility information provided by the infrastructure controller as set out in [GE/RT8015](#), demonstrate that the safety performance of the system will not be reduced under normal ('as designed') and defined degraded conditions of the infrastructure throughout the period that the train formation will operate on the **route**. This shall take into account normal, degraded and foreseeable fault conditions of the vehicle and shall consider abnormal situations such as assisting of a failed train.

C2 Infrastructure

C2.1

With the introduction of new or modified infrastructure, the infrastructure controller shall demonstrate compatibility of new or modified infrastructure systems with the vehicles using the **route** as set out in sections C2.2 to C2.4 below.

C2.2

The infrastructure controller shall demonstrate compatibility of the new or modified infrastructure with the vehicle systems, using the information provided by the train operator, and the methodology for demonstrating that compatibility as set out in Railway Group Standard [GE/RT8015](#).

C2.3

The infrastructure controller shall demonstrate that the safety performance of infrastructure systems will not be compromised by electromagnetic interference from the vehicle at the levels declared by the train operator.

C2.4

The infrastructure controller shall, using the vehicle susceptibility information provided by the train operator, as set out in [GE/RT8015](#), demonstrate that the safety performance of the train-borne system will not be reduced under normal ('as designed') degraded and foreseeable fault conditions of the vehicle throughout the period that the train formation is intended to operate on the **route**. This shall take into account normal, degraded and foreseeable fault conditions on the infrastructure.

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References

Railway Group Standards and other Railway Group Documents

GA/RT6001	Railway Group Standards Change Procedures
GA/RT6004	Temporary Non-Compliance with Railway Group Standards
GA/RT6006	Derogations from Railway Group Standards
GC/RT5212	Requirements for Defining and Maintaining Clearances
GE/RT8006	Interface between Rail Vehicle Weights and Underline Bridges
GE/RT8012	Controlling the Speed of Tilting Trains Through Curves
GE/RT8015	Electromagnetic Compatibility Between Railway Infrastructure and Trains
GE/RT8019	Tilting Trains: Controlling Tilt Systems to Maintain Clearances
GE/RT8023	Compatibility Between Electric Trains and Electrification Systems
GE/RT8025	Electrical Protective Provisions for Electrified Lines
GM/RT2000	Engineering Acceptance of Rail Vehicles
GM/RT2149	Requirements for Defining and Maintaining the Size of Railway Vehicles
GM/RT2453	Registration and Mandatory Data for Rail Vehicles
GM/TT0088	Permissible Track Forces for Railway Vehicles

The Catalogue of Railway Group Standards and the Railway Group Standards CD-ROM give the current issue number and status of documents published by Railway Safety.

Other References

Railway (Safety Case) Regulations 2000 including 2001 amendments

The Railways (Interoperability) (High-Speed) Regulations 2002

96/48/EC Council directive on the interoperability of the trans-European high speed rail system

2001/16/EC Council directive on the interoperability of the trans-European conventional rail system